Advanced Life Insurance Mathematics: Assignment II

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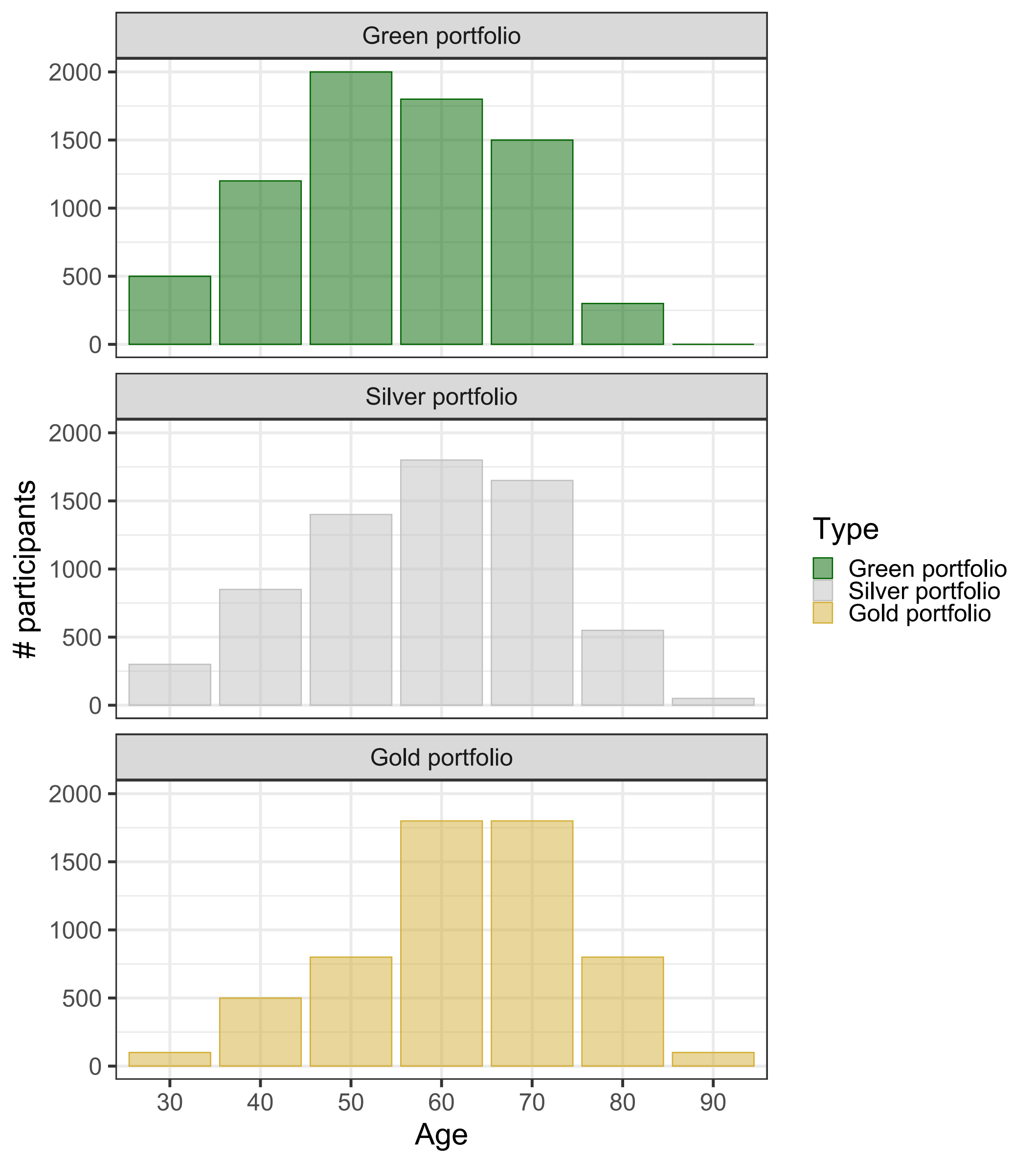
You work as an actuary for insurance company 'ISSurance', located in country 'XYZ' (*).

The CRO of the company is counting on you to assist with an urgent mission regarding the valuation of the liabilities of three pension plan portfolios.

The portfolios are categorized as a green portfolio, a silver and a golden portfolio. Their age composition is shown on the right.

Before Monday May, 18 you have to deliver a report with insights on the value of these liabilities. Your CRO prefers graphs over tables; keep this in mind during your mission.

Are you in for a challenge? Off we go!



(*) We leave it up to you (or your team) to decide on the location of country 'XYZ'.

On **phase 1** of your mission you decide to explore the historical evolution of mortality in country 'XYZ'.

Moreover, you want to be able to value a few basic concepts along period thinking.

You want to be able to

- deliver some insightful graphs on this evolution
- fit a (basic) stochastic mortality model (e.g., Lee-Carter) on the mortality data (for the scenario analysis you intend to run)
- evaluate the period life expectancy for a participant in the pension plan
- determine the expected present value (EPV) of a whole life annuity, issued to a participant in the plan.

(*) Check template functions `kannisto`, `LE_period` (for period LE) and `LA_period` (for the life annuity EPV).

Some considerations to keep in mind:

- What is the ultimate age you want to consider? How will you close (e.g.) the calibrated mortality rates for old ages?

- The EPV of a whole life annuity issued on (x) in period t, with annual payout of 10 000 EUR (in advance):

$$\ddot{a}^{(\text{per})}_{x,t} = 10\,000 \sum_{T=0}^{\infty} v(t,T)^T \, {}_T p_{x,t}$$

$${}_T p_{x,t} = \prod_{j=0}^{T-1} p_{x+j,t}, \quad v(t,T) = \frac{1}{1+i(t,T)}.$$

Here, ${}_T p_{x,t}$ is the T-year survival probability (period approach) at age x and time t and $v(t,T)$ is the annual discount factor at time t for time-to-maturity T.

Use the risk-free interest rate curve with volatility adjustment from EIOPA, available here.

On **phase 2** of your mission you step from period to cohort thinking. You focus on the best estimates for the future mortality rates obtained with the stochastic mortality model constructed in phase 1.

You want to be able to

- evaluate the cohort life expectancy for a participant in the pension plan (*)
- determine the expected present value (EPV) of a whole life annuity, issued to a participant in the plan, when applying cohort thinking.

Moreover, you want to be able to re-do these valuations under stress scenarios.

Can you come up with relevant shocks on the mortality and discount rates?

(*) Modify the functions `LE_period` and `LA_period` to a cohort setting.

Some considerations to keep in mind:

- the Solvency II longevity shock multiplies all future mortality rates with a factor 0.8
- the Solvency II mortality shock multiplies all future mortality rates with a factor 1.15
- EIOPA discount curves from multiple years are available (see previous RFR releases here); examine their impact on your valuations (e.g., 2019 versus 2026 release).

In for a challenge to earn bonus points?

- Examine the impact of interest rates on the EPV of a whole life annuity by generating different interest rate scenarios from the Vasicek Interest Rate Model, see here

On **phase 3** of your mission you bring in stochastics and decide to construct (say, 100) realistic scenarios for the evolution of future mortality rates.

You want to be able to

- evaluate the cohort life expectancy using each generated mortality scenario

- value the expected present value of a whole life annuity based on each generated mortality scenario.

Additionally, you want to assess how the phase 2 stress scenario valuations relate to your constructed EPV scenario valuations.

Some considerations to keep in mind:

- close each generated scenario for old age mortality
- visualize the distribution (e.g., histogram or density plot) of the simulated values for the cohort life expectancy
- same for the EPV of a whole life annuity (with cohort thinking).

On **phase 4** of your mission you are ready to value the cashflows and the liabilities of the three pension portfolios.

You want to be able to

- value the cash flows of a life long pension liability for a participant in the portfolio using the phase 2's best-estimate projections (*)
- same but now using the generated mortality scenarios from phase 3
- value the total portfolio cashflows over time.

Moreover, your CRO requires a best estimate of the present value of the total liabilities wrt each portfolio, the uncertainty of this estimate, and some useful insights in the differences across the three portfolios.

(*) Modify the function `LA_period`.

Some considerations to keep in mind:

- What is the retirement age you want to consider?
- The present value (in year 2026) of a cashflow in year 2026 + T for a life long pension liability is:

$$\sum_{c \in \mathcal{C}} I\left(x_c + T \geq 65\right) \cdot E_c \cdot {}_T p_{x_c,2026} \cdot v(2026, T)^T,$$

with $\mathcal{C}$ the different age classes in the portfolio, E_c the number of participants in age class c, 65 the chosen retirement age, and $I(\cdot)$ the indicator function.

In for a challenge to earn bonus points?

- Examine the impact of the stress scenarios (longevity and mortality shock, interest rate sensitivity) on the portfolio valuations.

On **phase 5** you put it all together!

```r
### Fit the method of Kannisto to close force of mortality at old ages
# mhat:       Matrix of death rates/forces of mortality (years, ages) - output fit701 function.
# est.ages:   The vector of ages used for estimating the parameters in the Kannisto logit model.
# proj.ages:  The vector of ages for which death rates/forces of mortality are to be projected/extrapolated.

kannisto ← function(mhat, est.ages, proj.ages) {
  # Extract years from rownames in 'mhat'
  years ← ...

  # Matrix to store closed mhat
  mhat.proj ← matrix(NA, nrow = nrow(mhat), ncol = length(proj.ages), dimnames = list(rownames(mhat), proj.ages))

  # For loop over each year
  for (t in ... ){
    # OLS estimation
    mhat.est       ← ...
    logit.mhat.est ← ...
    ols.est        ← ...

    # Extrapolate mhat
    phi1 ← ...
    phi2 ← ...
    mhat.proj[as.character(t),] ← ...
  }

  cbind(mhat, mhat.proj)
}
```

The template function `kannisto` makes use of the `mhat` attribute in the `fit701` object. This attribute stores the estimated forces of mortality as obtained from the calibrated Lee-Carter model.

```
str(LCfit701$mhat)
##  num [1:50, 1:91] 0.01074 0.01044 0.01043 0.00958 0.00891 ...
##  - attr(*, "dimnames")=List of 2
##   ..$ : chr [1:50] "1970" "1971" "1972" "1973" ...
##   ..$ : chr [1:91] "0" "1" "2" "3" ...
```

The `mhat` object has dimension names (`dimnames`): the row names are the years in the calibration period and the column names are the ages in the considered age range. Using these row and column names, you can easily extract information from `mhat`. E.g., you can retrieve the forces of mortality in the year 2019 for ages 80-90 as follows:

```
LCfit701$mhat[as.character(2019),as.character(80:90)]
##         80         81         82         83         84         85         86
## 0.03684590 0.04216823 0.04883158 0.05635619 0.06422193 0.07464614 0.08729304
##         87         88         89         90
## 0.10067436 0.11459144 0.13225372 0.15373687
```

```r
### Calculate period life expectancy for a particular age and year
# mhat: Matrix of (closed) death rates/forces of mortality (years, ages)
# Age:  Age at which to calculate period LE.
# Year: Year at which to calculate period LE.

LE_period ← function(mhat, Age, Year){
  # Extract years and ages from row and col names mhat
  years ← as.integer( ... )
  ages  ← as.integer( ... )

  # Extract forces of mortality in year 'Year' at ages 'Age' and beyond
  mtx   ← mhat[ ... , ... ]

  # First term in formula
  term1 ← (1 - exp(- ... ))/ ...

  # For loop to calculate summation term in formula
  term2 ← 0
  for(k in 1:( ... )){
    t1  ← prod(exp(- ... ))
    t2  ← (1 - exp(- ... ))/ ...
    term2 ← term2 +  ...
  }

  # Return period LE
   ...
}
```

```r
### Calculate EPV of a whole life annuity
# mhat:   Matrix of (closed) death rates/forces of mortality (years, ages)
# Age:    Age at which to value whole life annuity.
# Year:   Year at which to value whole life annuity.
# amount: Annual pay-out of the whole life annuity (in advance).
# rfr:    Risk-free-interest rate curve (column 'T': time to maturity, column 'i': interest rate).

LA_period <- function(mhat, Age, Year, amount, rfr){
  # Extract years and ages from row and col names mhat
  years <- as.integer( ... )
  ages  <- as.integer( ... )

  # One-year survival probabilities in year 'Year' for ages 'Age' and beyond
  pxt <- exp(-mhat[ ... , ... ])

  # T-year survival probabilities at age 'Age'
  Tpxt <- cumprod( ... )

  # Vector of discount factors
  v <-  ...

  # Maximum remaining years to live
  n <- tail(ages,1) - Age + 1

  # EPV
  amount  + amount * sum(sapply(1:n, function(j)  ... ))
}
```

Portfolio composition

Each portfolio consists solely of lifelong pension benefits, which are yearly payments starting from retirement (e.g., age 65) until death. For simplicity, the ages in the portfolio are rounded to the nearest decade, forming the age classes 30, 40, 50, 60, 70, 80, and 90. In the case study, you may assume that, e.g., participants in age group 30 are exactly 30 years old. The annual payment amount is 1, payable at the start of each year from age 65 onward, contingent upon the survival of the participant.

The table displays the number of participants across different age groups. For instance, the green portfolio has 500 participants in the 30-year-old age group and 1200 participants in the 40-year-old age group. The green portfolio tends to be more balanced towards younger ages, whereas the golden portfolio is more balanced towards older ages.

	Age 30	**40**	**50**	**60**	**70**	**80**	**90**
Green	500	1200	2000	1800	1500	300	0
Silver	300	850	1400	1800	1650	550	50
Gold	100	500	800	1800	1800	800	100


Good luck!
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